VERIFIED DATA CATALOG

Nine Accepted Negative Questions

Original and Mutated Pairs

Two-agent negative-generation workflow

This catalog renders the nine candidates accepted by Agent B. Each entry shows the complete original positive question, the complete accepted negative question, and the exact single-component replacement. Source 9 is absent because its candidate was verified false but rejected as a duplicate.

Example	Source	Mutation	Topic	Novelty
1	1	Assumption	Privacy Profiles	NOVEL BUT RELATED
2	2	Conclusion	Randomized Hyperparameter Tuning	NOVEL BUT RELATED
3	3	Algorithm	Correlated Gaussian Noise	NOVEL
4	4	Algorithm	Correlated Gaussian Noise	NOVEL BUT RELATED
5	5	Assumption	f-Differential Privacy	NOVEL
6	6	Conclusion	Gaussian Differential Privacy	NOVEL
7	7	Conclusion	Concentrated Differential Privacy	NOVEL BUT RELATED
8	8	Assumption	Private Geometric Median	NOVEL
9	10	Algorithm	l2 Mechanism	NOVEL BUT RELATED

TRUE denotes the original accepted positive question. FALSE denotes the verified negative mutation.

Example 1: Source 1

Mutation	Assumption	**Novelty**	NOVEL BUT RELATED
Candidate	`newneg_source_1_assumption_1`	**Proof mode**	Explicit Counterexample

TRUE **Original positive question**

Let $X \sim X'$ denote neighboring datasets under a fixed add/remove or replace-one adjacency relation. Let $m \in \mathbb{N}$, $[m] = \{1, \dots, m\}$, and let f_i be real-valued scores satisfying $|f_i(X) - f_i(X')| \le 1$ for every $i \in [m]$ and every $X \sim X'$. Let $Z_1, \dots, Z_m$ be i.i.d. real noises with common density p, and define $M(X) = \arg\max_{i \in [m]} \{f_i(X) + Z_i\}$ using fixed data-independent tie-breaking. For distributions P, Q with densities p_P, p_Q, define the hockey-stick divergence $H_{e^t}(P\|Q) = \int [p_P(y)/p_Q(y) - e^t]_+ p_Q(y)\, dy$, where $[u]_+ = \max\{u, 0\}$. Define $\delta_2(t) = \sup_{|a-a'| \le 2} H_{e^t}(\mathrm{Law}(a + Z_1) \| \mathrm{Law}(a' + Z_1))$. Is it true that, for every $t \in \mathbb{R}$ and every $X \sim X'$, one has $H_{e^t}(M(X)\|M(X')) \le m\, \delta_2(t)$?

FALSE **Accepted negative question**

Let $X \sim X'$ denote neighboring datasets under a fixed add/remove or replace-one adjacency relation. Let $m \in \mathbb{N}$, $[m] = \{1, \dots, m\}$, and let f_i be real-valued scores satisfying $|f_i(X) - f_i(X')| \le 1$ for every $i \in [m]$ and every $X \sim X'$. Let $Z_1, \dots, Z_m$ be identically distributed real noises with common density p, and define $M(X) = \arg\max_{i \in [m]} \{f_i(X) + Z_i\}$ using fixed data-independent tie-breaking. For distributions P, Q with densities p_P, p_Q, define the hockey-stick divergence $H_{e^t}(P\|Q) = \int [p_P(y)/p_Q(y) - e^t]_+ p_Q(y)\, dy$, where $[u]_+ = \max\{u, 0\}$. Define $\delta_2(t) = \sup_{|a-a'| \le 2} H_{e^t}(\mathrm{Law}(a + Z_1) \| \mathrm{Law}(a' + Z_1))$. Is it true that, for every $t \in \mathbb{R}$ and every $X \sim X'$, one has $H_{e^t}(M(X)\|M(X')) \le m\, \delta_2(t)$?

Exact single-component mutation

Before	i.i.d. real noises
After	identically distributed real noises

Source: Koskela, Redberg, and Wang (2024), Privacy Profiles for Private Selection. Agent B: FALSE, VERIFIED, ACCEPT.

Example 2: Source 2

Mutation	Conclusion	**Novelty**	NOVEL BUT RELATED
Candidate	`newneg_source_2_conclusion_1`	**Proof mode**	Exact Privacy Loss Calculation

TRUE **Original positive question**

Fix $\gamma \in (0,1)$ and $\eta \in (-1,\infty)$. Let $K \in \mathbb{N}$ have the truncated negative-binomial law $D_{\eta,\gamma}$ defined, for $k \in \mathbb{N}$, by $\Pr(K=k) = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1}\prod_{i=0}^{k-1}\frac{i+\eta}{i+1}$ when $\eta \neq 0$, and by $\Pr(K=k) = \frac{(1-\gamma)^k}{k\log(1/\gamma)}$ when $\eta = 0$. A selection algorithm A draws K independently, runs a base mechanism Q on the same dataset K times, and releases the run with the best value in a fixed ordered quality-score space. Under the paper's neighboring-dataset relation, suppose Q is ε-differentially private, meaning $\Pr(Q(X) \in E) \le e^{\varepsilon}\Pr(Q(X') \in E)$ for all neighboring X, X' and measurable E. Is A necessarily $(\eta+2)\varepsilon$-differentially private?

FALSE **Accepted negative question**

Fix $\gamma \in (0,1)$ and $\eta \in (-1,\infty)$. Let $K \in \mathbb{N}$ have the truncated negative-binomial law $D_{\eta,\gamma}$ defined, for $k \in \mathbb{N}$, by $\Pr(K=k) = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1}\prod_{i=0}^{k-1}\frac{i+\eta}{i+1}$ when $\eta \neq 0$, and by $\Pr(K=k) = \frac{(1-\gamma)^k}{k\log(1/\gamma)}$ when $\eta = 0$. A selection algorithm A draws K independently, runs a base mechanism Q on the same dataset K times, and releases the run with the best value in a fixed ordered quality-score space. Under the paper's neighboring-dataset relation, suppose Q is ε-differentially private, meaning $\Pr(Q(X) \in E) \le e^{\varepsilon}\Pr(Q(X') \in E)$ for all neighboring X, X' and measurable E. Is A necessarily $(\eta+1)\varepsilon$-differentially private?

Exact single-component mutation

Before $(\eta+2)\varepsilon$-differentially private

After $(\eta+1)\varepsilon$-differentially private

Source: Koskela, Redberg, and Wang (2024), Privacy Profiles for Private Selection. Agent B: FALSE, VERIFIED, ACCEPT.

Example 3: Source 3

Mutation	Algorithm	**Novelty**	NOVEL
Candidate	`newneg_source_3_algorithm_1`	**Proof mode**	Exact Variance Calculation

TRUE **Original positive question**

Let $d, n \in \mathbb{N}$, $\mu > 0$, $X = (x_1, \ldots, x_n)$ with $x_i \in [0,1]^d$, and $f(X) = \sum_{i=1}^n x_i$. Let $\mathbf{1}_d$ be the all-ones vector in $\mathbb{R}^d$. Independently sample $\eta \sim \mathcal{N}(0, (\sqrt{d}+1)/(4\mu^2))$ and $Z \sim \mathcal{N}(0, ((d+\sqrt{d})/(4\mu^2))I_d)$, where I_d is the $d \times d$ identity. If the released query vector is $\widetilde{f} = f(X) + \eta \mathbf{1}_d + Z$, is the error in every coordinate distributed as $\mathcal{N}(0, (d + 2\sqrt{d} + 1)/(4\mu^2))$, with standard deviation $(\sqrt{d}+1)/(2\mu)$?

FALSE **Accepted negative question**

Let $d, n \in \mathbb{N}$, $\mu > 0$, $X = (x_1, \ldots, x_n)$ with $x_i \in [0,1]^d$, and $f(X) = \sum_{i=1}^n x_i$. Let $\mathbf{1}_d$ be the all-ones vector in $\mathbb{R}^d$. Independently sample $\eta \sim \mathcal{N}(0, (\sqrt{d}+1)/(4\mu^2))$ and $Z \sim \mathcal{N}(0, ((d-\sqrt{d})/(4\mu^2))I_d)$, where I_d is the $d \times d$ identity. If the released query vector is $\widetilde{f} = f(X) + \eta \mathbf{1}_d + Z$, is the error in every coordinate distributed as $\mathcal{N}(0, (d + 2\sqrt{d} + 1)/(4\mu^2))$, with standard deviation $(\sqrt{d}+1)/(2\mu)$?

Exact single-component mutation

Before $Z \sim \mathcal{N}(0, ((d+\sqrt{d})/(4\mu^2))I_d)$

After $Z \sim \mathcal{N}(0, ((d-\sqrt{d})/(4\mu^2))I_d)$

Source: Lebeda (2025), Better Gaussian Mechanism using Correlated Noise. Agent B: FALSE, VERIFIED, ACCEPT.

Example 4: Source 4

Mutation	Algorithm	**Novelty**	NOVEL BUT RELATED
Candidate	`newneg_source_4_algorithm_1`	**Proof mode**	Exact Gaussian Tradeoff Counterexample

TRUE **Original positive question**

Let $d, n \in \mathbb{N}$, $\mu > 0$, $X = (x_1, \ldots, x_n)$ with $x_i \in [0,1]^d$, $f(X) = \sum_{i=1}^n x_i$, and let $\mathbf{1}_d$ and I_d denote the all-ones vector and identity matrix. Two datasets are add/remove neighbors if one is obtained from the other by adding or removing one row. Independently sample $\eta \sim \mathcal{N}(0, (\sqrt{d}+1)/(4\mu^2))$ and $Z \sim \mathcal{N}(0, ((d+\sqrt{d})/(4\mu^2))I_d)$, and release $A(X) = (f(X) + \eta \mathbf{1}_d + Z,\ n + 2\eta)$. For distributions P, Q, let $T(P, Q)$ be their hypothesis-testing trade-off function; a mechanism is μ-GDP if $T(A(X), A(X')) \geq T(\mathcal{N}(0,1), \mathcal{N}(\mu, 1))$ pointwise for every neighboring X, X'. Does A satisfy μ-GDP under add/remove adjacency?

FALSE **Accepted negative question**

Let $d, n \in \mathbb{N}$, $\mu > 0$, $X = (x_1, \ldots, x_n)$ with $x_i \in [0,1]^d$, $f(X) = \sum_{i=1}^n x_i$, and let $\mathbf{1}_d$ and I_d denote the all-ones vector and identity matrix. Two datasets are add/remove neighbors if one is obtained from the other by adding or removing one row. Independently sample $\eta \sim \mathcal{N}(0, (\sqrt{d}+1)/(4\mu^2))$ and $Z \sim \mathcal{N}(0, ((d+\sqrt{d})/(4\mu^2))I_d)$, and release $A(X) = (f(X) + \eta \mathbf{1}_d + Z,\ n + \eta)$. For distributions P, Q, let $T(P, Q)$ be their hypothesis-testing trade-off function; a mechanism is μ-GDP if $T(A(X), A(X')) \geq T(\mathcal{N}(0,1), \mathcal{N}(\mu, 1))$ pointwise for every neighboring X, X'. Does A satisfy μ-GDP under add/remove adjacency?

Exact single-component mutation

Before $A(X) = (f(X) + \eta \mathbf{1}_d + Z,\ n + 2\eta)$

After $A(X) = (f(X) + \eta \mathbf{1}_d + Z,\ n + \eta)$

Source: Lebeda (2025), Better Gaussian Mechanism using Correlated Noise. Agent B: FALSE, VERIFIED, ACCEPT.

Example 5: Source 5

Mutation	Assumption	**Novelty**	NOVEL
Candidate	`newneg_source_5_assumption_1`	**Proof mode**	Explicit Tradeoff Counterexample

TRUE **Original positive question**

Let S, S^- be remove-neighboring datasets, and let $M(S)$ be a distribution on an output space. For distributions P, Q, define the trade-off function $T[P, Q](\alpha) = \inf_\phi\{1 - \mathbb{E}_Q[\phi] : \mathbb{E}_P[\phi] \le \alpha\}$, where the infimum ranges over measurable randomized tests ϕ taking values in $[0, 1]$. A mechanism is f-DP if both $T[M(S), M(S^-)] \ge f$ and $T[M(S^-), M(S)] \ge f$ pointwise for every remove pair. For $\varepsilon \ge 0$ and $\delta \in [0, 1]$, define $f_{\varepsilon,\delta}(\alpha) = \max\{0, 1 - \delta - e^\varepsilon\alpha, e^{-\varepsilon}(1 - \delta - \alpha)\}$. Is M (ε, δ)-DP in both directions for every remove pair if and only if it is $f_{\varepsilon,\delta}$-DP?

FALSE **Accepted negative question**

Let S, S^- be remove-neighboring datasets, and let $M(S)$ be a distribution on an output space. For distributions P, Q, define the trade-off function $T[P, Q](\alpha) = \inf_\phi\{1 - \mathbb{E}_Q[\phi] : \mathbb{E}_P[\phi] \le \alpha\}$, where the infimum ranges over measurable randomized tests ϕ taking values in $[0, 1]$. A mechanism is f-DP if both $T[M(S), M(S^-)] \ge f$ and $T[M(S^-), M(S)] \ge f$ pointwise for every remove pair. For $\varepsilon \ge 0$ and $\delta \in [0, 1]$, define $f_{\varepsilon,\delta}(\alpha) = \max\{0, 1 - \delta - e^\varepsilon\alpha, e^{\varepsilon}(1 - \delta - \alpha)\}$. Is M (ε, δ)-DP in both directions for every remove pair if and only if it is $f_{\varepsilon,\delta}$-DP?

Exact single-component mutation

Before $f_{\varepsilon,\delta}(\alpha) = \max\{0, 1 - \delta - e^\varepsilon\alpha, e^{-\varepsilon}(1 - \delta - \alpha)\}$

After $f_{\varepsilon,\delta}(\alpha) = \max\{0, 1 - \delta - e^\varepsilon\alpha, e^{\varepsilon}(1 - \delta - \alpha)\}$

Source: Tran, Koskela, Räisä, and Honkela (2026), f-Differential Privacy Filters: Validity and Approximate Solutions. Agent B: FALSE, VERIFIED, ACCEPT.

Example 6: Source 6

Mutation	Conclusion	**Novelty**	NOVEL
Candidate	`newneg_source_6_conclusion_1`	**Proof mode**	Exact Gaussian Tradeoff Calculation

TRUE **Original positive question**

For distributions P, Q, let $T[P, Q]$ denote their hypothesis-testing trade-off function. For $\mu \geq 0$, define the Gaussian trade-off function $G_\mu = T[\mathcal{N}(0,1), \mathcal{N}(\mu,1)]$. For trade-off functions represented as $f_1 = T[P_1, Q_1]$ and $f_2 = T[P_2, Q_2]$, define their tensor product by $f_1 \otimes f_2 = T[P_1 \times P_2, Q_1 \times Q_2]$. For every $\mu_1, \mu_2 \geq 0$, is $G_{\mu_1} \otimes G_{\mu_2} = G_{\sqrt{\mu_1^2+\mu_2^2}}$?

FALSE **Accepted negative question**

For distributions P, Q, let $T[P, Q]$ denote their hypothesis-testing trade-off function. For $\mu \geq 0$, define the Gaussian trade-off function $G_\mu = T[\mathcal{N}(0,1), \mathcal{N}(\mu,1)]$. For trade-off functions represented as $f_1 = T[P_1, Q_1]$ and $f_2 = T[P_2, Q_2]$, define their tensor product by $f_1 \otimes f_2 = T[P_1 \times P_2, Q_1 \times Q_2]$. For every $\mu_1, \mu_2 \geq 0$, is $G_{\mu_1} \otimes G_{\mu_2} = G_{\mu_1+\mu_2}$?

Exact single-component mutation

Before $G_{\sqrt{\mu_1^2+\mu_2^2}}$

After $G_{\mu_1+\mu_2}$

Source: Tran, Koskela, Räisä, and Honkela (2026), f-Differential Privacy Filters: Validity and Approximate Solutions. Agent B: FALSE, VERIFIED, ACCEPT.

Example 7: Source 7

Mutation	Conclusion	**Novelty**	NOVEL BUT RELATED
Candidate	`newneg_source_7_conclusion_1`	**Proof mode**	Explicit Gaussian Dp Counterexample

TRUE **Original positive question**

Let neighboring datasets use replacement adjacency. For $\alpha > 1$, define the Renyi divergence $D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\int (dP/dQ)^\alpha\, dQ$. Suppose a randomized mechanism A is ρ-zCDP for some $\rho \geq 0$, meaning that $D_\alpha(A(X)\|A(X')) \leq \rho\alpha$ for every $\alpha > 1$ and every neighboring X, X'. For every $\delta \in (0,1)$, is A necessarily $(\rho + 2\sqrt{\rho\log(1/\delta)}, \delta)$-differentially private?

FALSE **Accepted negative question**

Let neighboring datasets use replacement adjacency. For $\alpha > 1$, define the Renyi divergence $D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\int (dP/dQ)^\alpha\, dQ$. Suppose a randomized mechanism A is ρ-zCDP for some $\rho \geq 0$, meaning that $D_\alpha(A(X)\|A(X')) \leq \rho\alpha$ for every $\alpha > 1$ and every neighboring X, X'. For every $\delta \in (0,1)$, is A necessarily $(2\sqrt{\rho\log(1/\delta)}, \delta)$-differentially private?

Exact single-component mutation

Before $(\rho + 2\sqrt{\rho\log(1/\delta)}, \delta)$-differentially private

After $(2\sqrt{\rho\log(1/\delta)}, \delta)$-differentially private

Source: Haghifam, Steinke, and Ullman (2024), Private Geometric Median. Agent B: FALSE, VERIFIED, ACCEPT.

Example 8: Source 8

Mutation	Assumption	**Novelty**	NOVEL
Candidate	`newneg_source_8_assumption_1`	**Proof mode**	Explicit Sensitivity Counterexample

TRUE **Original positive question**

Let $n \in \mathbb{N}$, $X = (x_1, \ldots, x_n) \in (\mathbb{R}^d)^n$, $\gamma \in [1/2, 1]$, $m = \lceil \gamma n \rceil$, and $\nu \geq 0$. Define $N_i^X(\nu) = |\{j \in [n] : \|x_j - x_i\|_2 \leq \nu\}|$ and $N_X(\nu) = \frac{1}{m} \max_{I \subseteq [n],\, |I|=m} \sum_{i \in I} N_i^X(\nu)$. If replacement-neighboring datasets X, X' differ in one sample, is $|N_X(\nu) - N_{X'}(\nu)| \leq 3$?

FALSE **Accepted negative question**

Let $n \in \mathbb{N}$, $X = (x_1, \ldots, x_n) \in (\mathbb{R}^d)^n$, $\gamma \in [1/3, 1]$, $m = \lceil \gamma n \rceil$, and $\nu \geq 0$. Define $N_i^X(\nu) = |\{j \in [n] : \|x_j - x_i\|_2 \leq \nu\}|$ and $N_X(\nu) = \frac{1}{m} \max_{I \subseteq [n],\, |I|=m} \sum_{i \in I} N_i^X(\nu)$. If replacement-neighboring datasets X, X' differ in one sample, is $|N_X(\nu) - N_{X'}(\nu)| \leq 3$?

Exact single-component mutation

Before $\gamma \in [1/2, 1]$

After $\gamma \in [1/3, 1]$

Source: Haghifam, Steinke, and Ullman (2024), Private Geometric Median. Agent B: FALSE, VERIFIED, ACCEPT.

Example 9: Source 10

Mutation	Algorithm	**Novelty**	NOVEL BUT RELATED
Candidate	`newneg_source_10_algorithm_1`	**Proof mode**	Exact Moment Calculation

TRUE **Original positive question**

Let $d \in \mathbb{N}$ and $\sigma > 0$. The d-dimensional ℓ_2 mechanism releases $T(X) + Y$, where $Y \in \mathbb{R}^d$ has density proportional to $\exp(-\|y\|_2/\sigma)$. Is its mean squared ℓ_2 error equal to $\mathbb{E}\|Y\|_2^2 = d(d+1)\sigma^2$?

FALSE **Accepted negative question**

Let $d \in \mathbb{N}$ and $\sigma > 0$. The d-dimensional ℓ_2 mechanism releases $T(X) + Y$, where $Y \in \mathbb{R}^d$ has density proportional to $\exp(-\|y\|_2^2/\sigma)$. Is its mean squared ℓ_2 error equal to $\mathbb{E}\|Y\|_2^2 = d(d+1)\sigma^2$?

Exact single-component mutation

Before $\exp(-\|y\|_2/\sigma)$

After $\exp(-\|y\|_2^2/\sigma)$

Source: Joseph, Kulesza, and Yu (2025), Approximate Differential Privacy of the l2 Mechanism. Agent B: FALSE, VERIFIED, ACCEPT.